OBJECT-ORIENTED PROGRAMMING / GROUP PROJECT

Hostel Maintenance Request Tracker

PART 1 PROPOSAL • Java + Javalin backend with a simple web front end

THE PROBLEM

Residents of Zambia University College Of Technology often report repairs informally, in person or by message. Requests get forgotten or reported twice, and residents cannot tell whether a request was received, who is handling it, or whether it has been fixed. This tracker gives residents and maintenance staff one shared record for every repair request.

INTENDED USERS

RESIDENTS

Submit a repair with location and description; check progress by reference number.

MAINTENANCE STAFF

View incoming requests, take responsibility for work, and update progress.

PROPOSED WORKFLOW AND SCOPE

A resident submits a request in a web form. The Java/Javalin backend validates and saves it, then returns a reference number. Maintenance staff view open requests and update status: Received → In Progress → Resolved. Residents check the latest status by reference number. Version one covers request recording, status tracking, and simple persistence (file or SQLite); it excludes emergency dispatch, chat, notifications, and facilities-management integrations.

MAIN CLASSES (6)

CLASS	RESPONSIBILITY
User	Base class for shared identity and contact details; defines the common user role.
Resident	A User with a room number and hostel block who submits repair requests and checks them by reference number.
MaintenanceStaff	A User who views incoming requests and updates work status.
MaintenanceRequest	Stores reference, location, description, resident, optional assigned staff, status, and timestamps.
RequestService	Validates submissions, coordinates assignment and updates, and enforces valid status changes.
RequestRepository	Saves, finds, and loads requests using one simple persistent store (file or SQLite).

Relationship: Resident and MaintenanceStaff inherit shared identity from User. Both interact with requests through RequestService, which uses RequestRepository for persistence.

TEAM (5 STUDENTS REQUIRED)

Tehilla Chiwama — 2410014

Mwenya Simpungwe — 2410013

Bright Mwansa — 2410010

Cheela Mulilo — 2410032

PENDING — fifth student's name and student number to be added (not yet provided).